Supervised Machine Learning
K-Nearest Neighbor

CSE4107: Artificial Intelligence

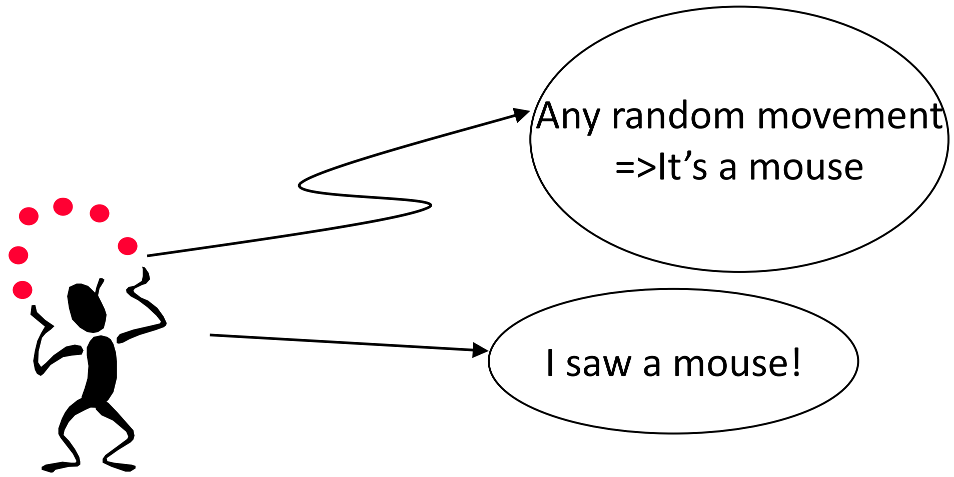
References

- https://codekarim.com/node/32
- https://towardsdatascience.com/how-to-find-the-optimal-value-of-k-in-knn-35d936e554eb#:~:text=The%20optimal%20K%20value%20usually,be%20aware%20of%20the%20outliers.

Eager Learning

Eager Learning: Eager learning methods construct general (one-fits-all), explicit (input dependent) description of the target function based on the provided training examples.

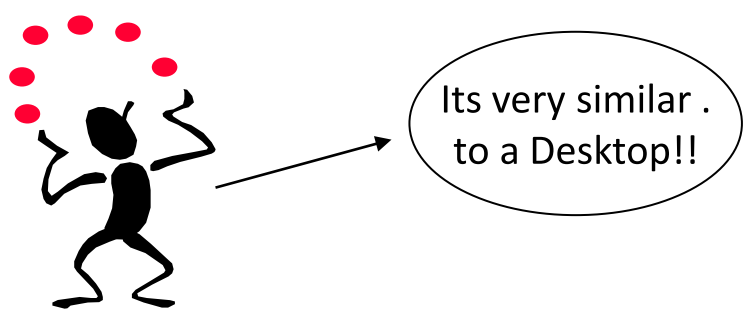
- Explicit description of target function on the whole training set
- Eager learning methods use the same approximation to the target function, which must be learned based on training examples and before input queries are observed.



Instance-based Learning

Instance-based Learning

- Memory-based Learning
- Instead of performing explicit generalization, compares new problem instances with instances seen in training, which have been stored in memory
- Constructs hypothesis directly from the training instances
- Referred to as "Lazy" learning
- **K-Nearest Neighbor**



K-Nearest Neighbor

- It is a simple algorithm that stores all available cases and classifies new cases based on a similarity measure (distance function).
- It has been used in statistical estimation and pattern recognition since 1970's.

Features

- All instances correspond to points in an n-dimensional Euclidean space
- Classification is delayed till a new instance arrives
- Classification done by comparing feature vectors of the different points
- Target function may be discrete or real-valued

K-Nearest Neighbor

- A case is classified by a majority voting of its neighbors, with the case being assigned to the class most common among its K nearest neighbors measured by a distance function.
- If K=1, then the case is simply assigned to the class of its nearest neighbor

K-Nearest Neighbor

Distance Function Measurements

- Euclidean Distance: $\sqrt{\sum_{i=1}^{k}(x_i - y_i)^2}$
- Manhattan Distance: $\sum_{i=1}^{k}|x_i - y_i|$
- Minkowski Distance: $\left(\sum_{i=1}^{k}(|x_i - y_i|)^q\right)^{\frac{1}{q}}$

K-Nearest Neighbor

- **Hamming Distance**

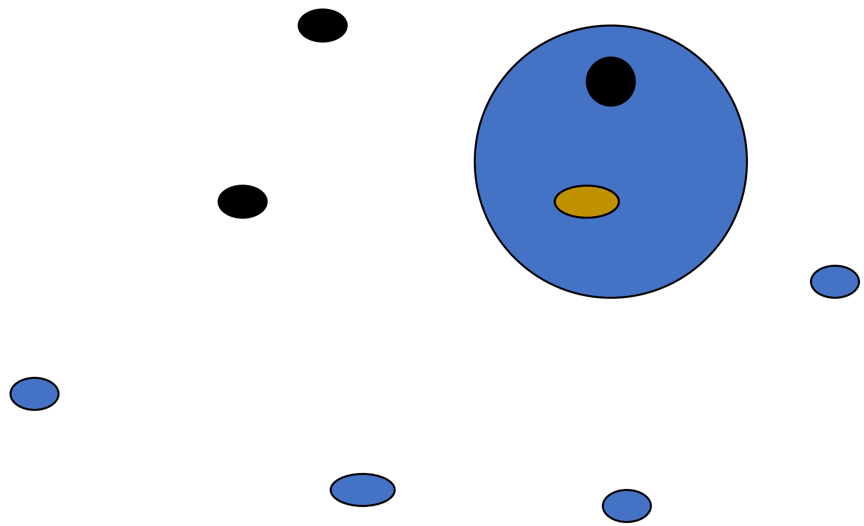
$$D_H = \sum_{i=1}^{k} |x_i - y_i|$$

- If $x = y$, then $D = 0$
- If $x \neq y, then\ D = 1$

X	Y	Distance
Male	Male	0
Male	Female	1

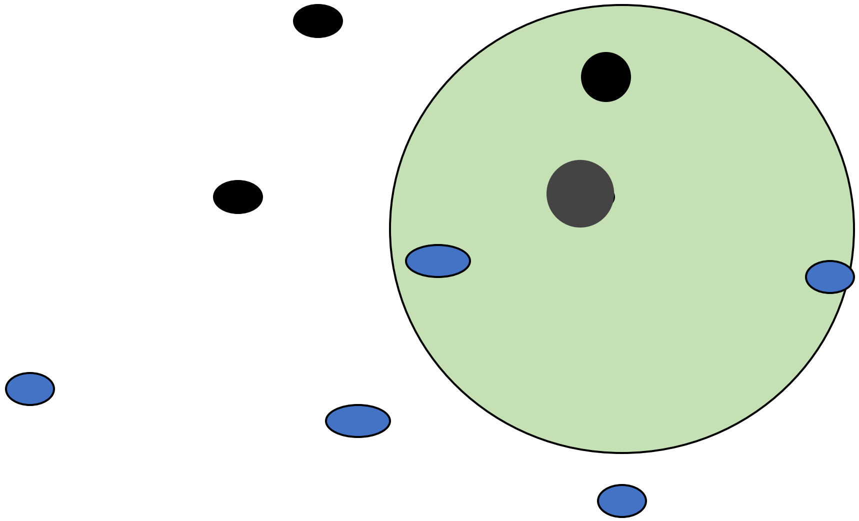
K-Nearest Neighbor

- **One Nearest Neighbor**



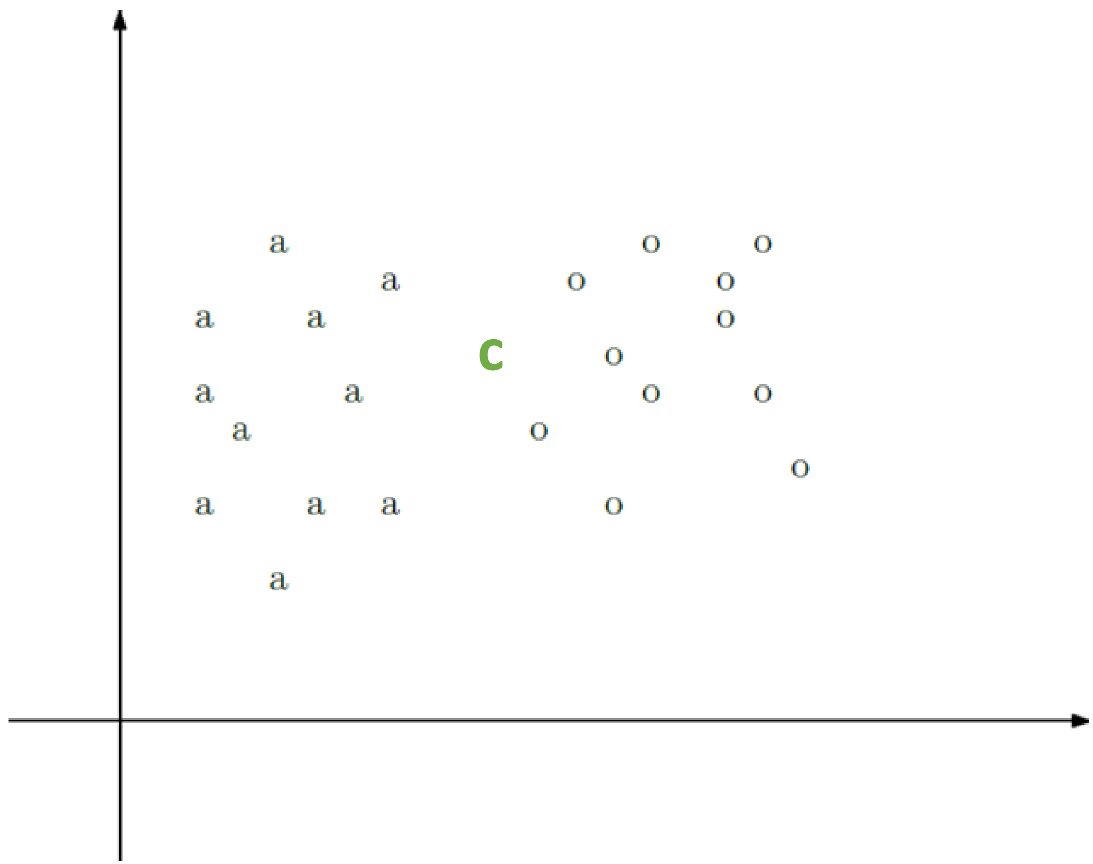
K-Nearest Neighbor

- **Three Nearest Neighbor**



K-Nearest Neighbor

- Selecting **"K"**

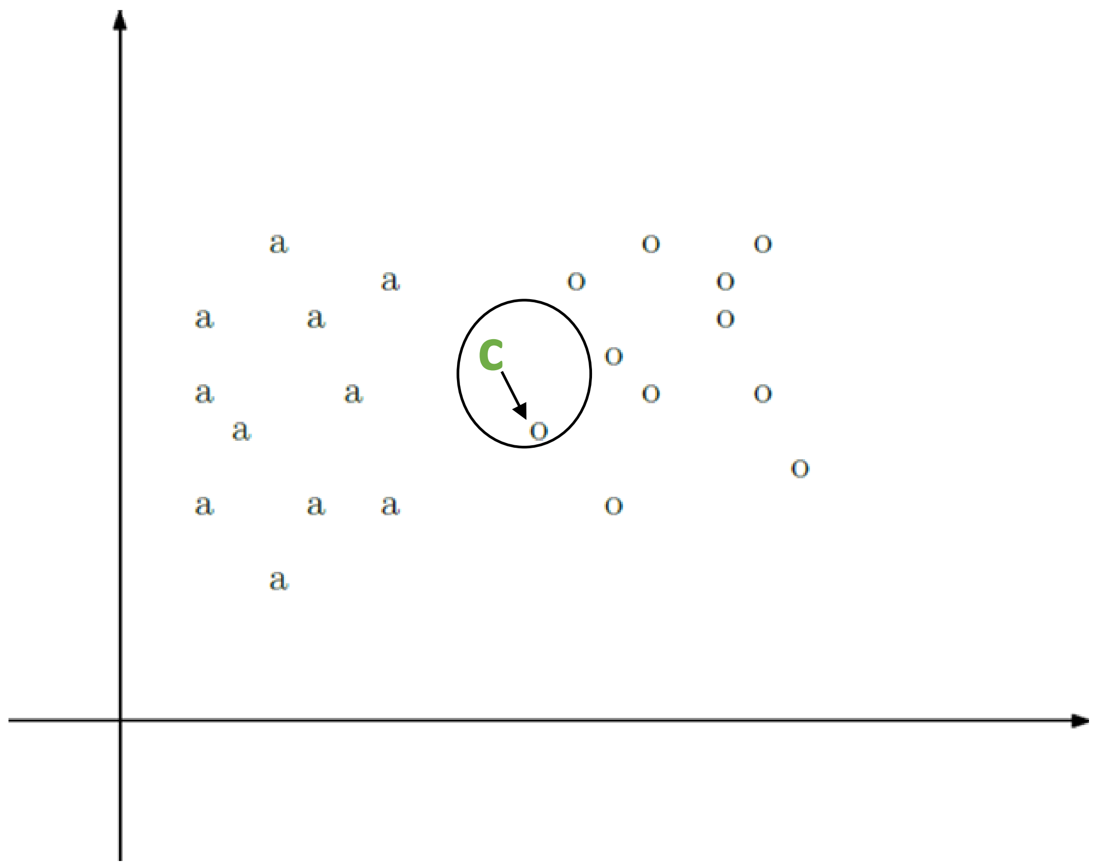


K-Nearest Neighbor

- Selecting **"K"**

If ***K = 1***

- *Class:* ***O***

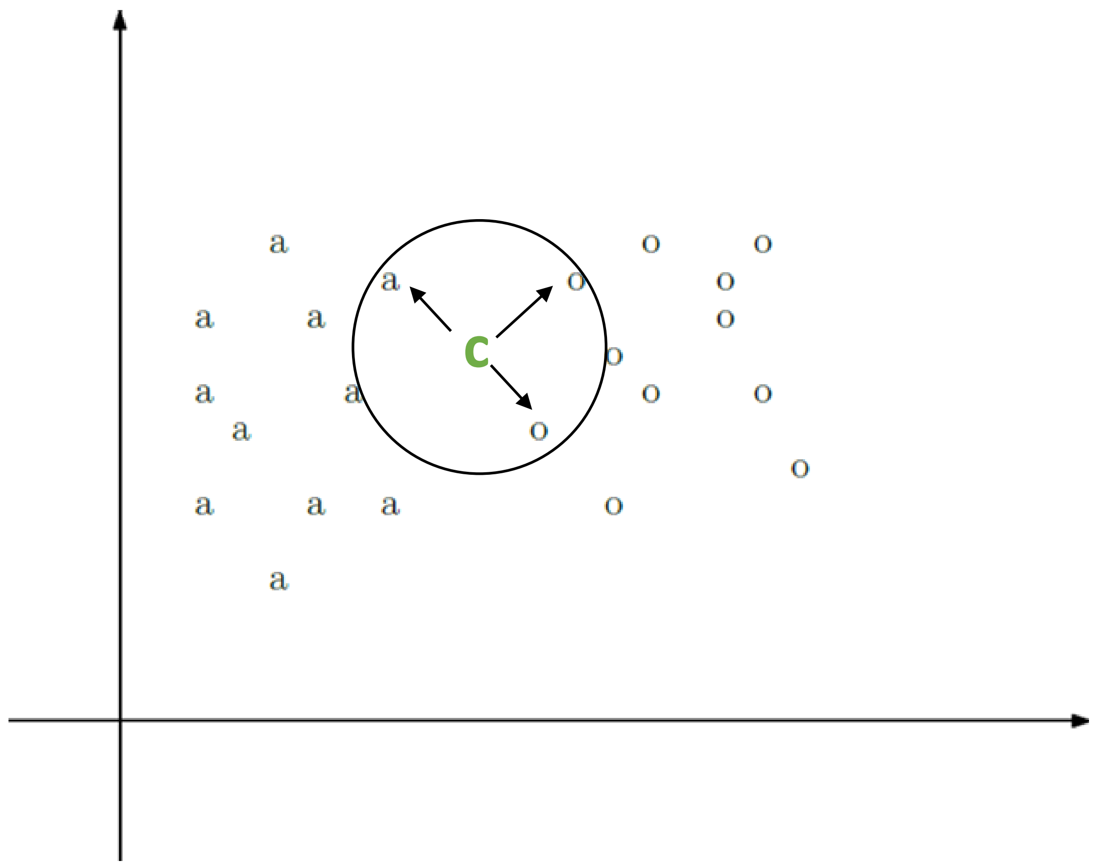


K-Nearest Neighbor

- Selecting **"K"**

If ***K = 3***

- *Class:* ***O***

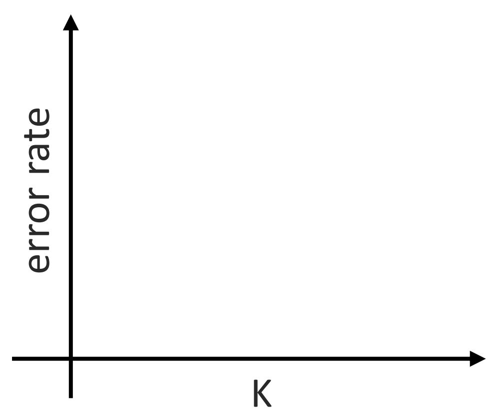


K-Nearest Neighbor

How to select an optimal value for ***K***?

- There are no pre-defined statistical methods to find the most favorable value of K
- Initialize a random K value and start computing
- Choosing a small value of K leads to unstable decision boundaries

- **Derive a plot between error rate and K denoting values in a defined range. Then choose the K value as having a minimum error rate.**



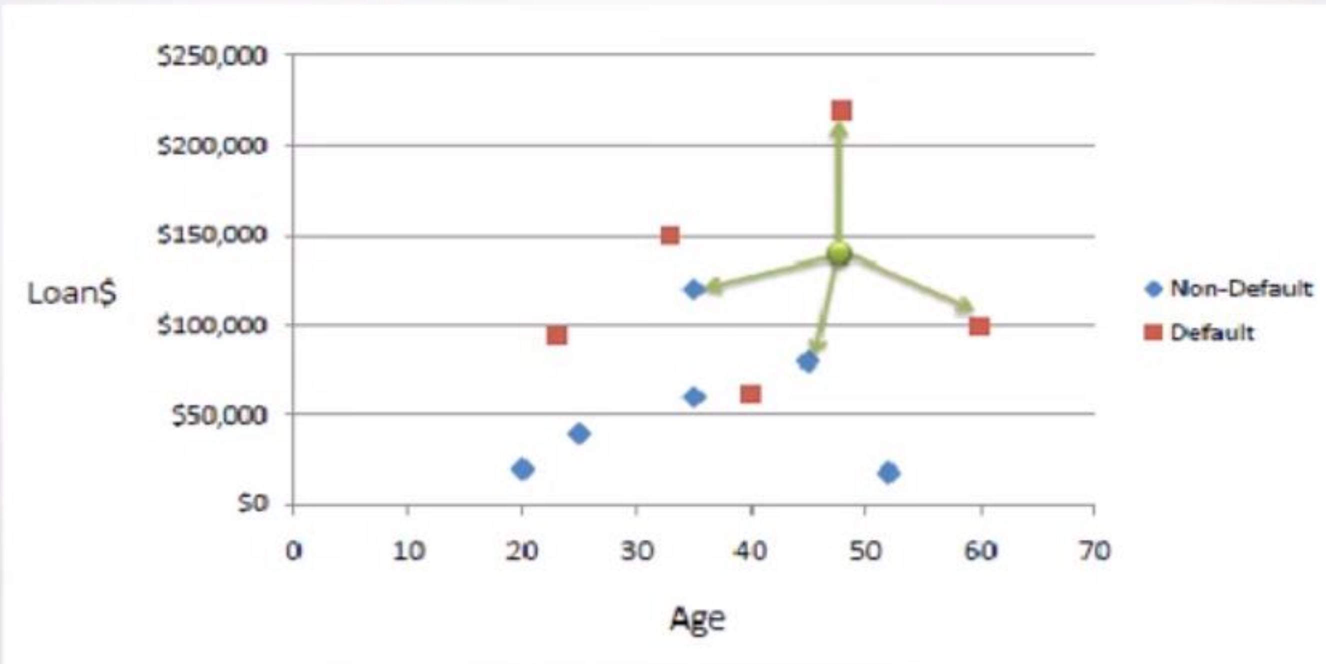
K-Nearest Neighbor

Working Flow

- Initialize the K value.
- Calculate the distance between test input and K trained nearest neighbors.
- Check class categories of nearest neighbors and determine the type in which test input falls.
- Classification will be done by taking the majority of votes.
- Return the class category.

K-Nearest Neighbor — Math

- Consider the following data concerning credit default. Age and Loan are two numerical variables (predictors) and Default is the target



K-Nearest Neighbor — Math

- We can now use the training set to classify an unknown case (Age=48 and Loan=$142,000) using Euclidean distance.
- If K=1 then the nearest neighbor is the last case in the training set with Default=Y.
- D = Sqrt[(48-33)^2 + (142000-150000)^2] = 8000.01 >> Default=Y

Age	Loan	Default	Distance	
25	$40,000	N	102000	
35	$60,000	N	82000	
45	$80,000	N	62000	
20	$20,000	N	122000	
35	$120,000	N	22000	2
52	$18,000	N	124000	
23	$95,000	Y	47000	
40	$62,000	Y	80000	
60	$100,000	Y	42000	3
48	$220,000	Y	78000	
33	$150,000	Y	8000	1
48	$142,000	?		

Euclidean Distance

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

- With K=3, there are two Default=Y and one Default=N out of three closest neighbors. The prediction for the unknown case is again Default=Y

K-Nearest Neighbor — Math

Normalization

- One major drawback in calculating distance measures directly from the training is in the case where variables have difference measurement scales or there is a mixture of numerical and categorical variables.
- For example, if one variable is based on annual income in dollars, and the other is based on age in years then income will have a much higher influence on the distance calculated.
- One solution is to standardized the training set.

K-Nearest Neighbor — Math

Normalization

$$X_i = \frac{X - Min}{Max - Min}$$

- Using the standardized distance on the same training set, the unknown case returned a different neighbor which is not a good sign of robustness

K-Nearest Neighbor — Math

Normalization

Age	Loan	Default	Distance
0.125	0.11	N	0.7652
0.375	0.21	N	0.5200
0.625	0.31	N	0.3160
0	0.01	N	0.9245
0.375	0.50	N	0.3428
0.8	0.00	N	0.6220
0.075	0.38	Y	0.6669
0.5	0.22	Y	0.4437
1	0.41	Y	0.3650
0.7	1.00	Y	0.3861
0.325	0.65	Y	0.3771
0.7	0.61	?	

Standardized Variable

$$X_s = \frac{X - Min}{Max - Min}$$

K-Nearest Neighbor

- Unlike all the previous learning methods, kNN does not build model from the training data.
- To classify a test instance *d*, define *k*-neighborhood *P* as *k* nearest neighbors of *d*
- Count number *n* of training instances in *P* that belong to class c_j
- Estimate $\Pr(c_j|d)$ as n/k
- No training is needed. Classification time is linear in training set size for each test case.

Discussion

- kNN can deal with complex and arbitrary decision boundaries
- Despite its simplicity, researchers have shown that the classification accuracy of kNN can be quite strong and in many cases as accurate as those elaborated methods
- kNN is slow at the classification time
- kNN does not produce an understandable model

Miscellaneous

- Voronoi Diagram
- Cosine Similarity
- Curse of Dimensionality